Do Now · retrieval

Writing, following and documenting algorithms

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Give a test for a range 0–100 that should be rejected.

Q2

What does return do?

Q3

What is the last index for an array of length 4?

Q4

Why is a[0] unsafe for an empty array?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

State the precondition — The array must not be empty, or a[0] cannot be read.

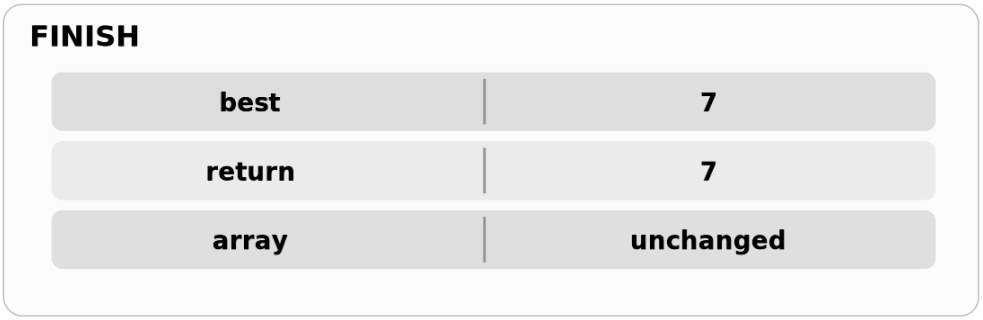
Core idea 2

Trace the maximum — Set best to 3; compare 7, update best to 7, then ignore 2 and 5.

Core idea 3

Document contract — Input: nonempty int array. Output: greatest element. Does not modify input.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

algorithm — Precise finite steps that solve a problem.

precondition — Requirement inputs must meet before execution.

postcondition — Condition guaranteed if the algorithm completes correctly.

invariant — Property maintained during repeated steps.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

Trace Largest([4,2,9]) and write the returned result.

Follow the worked method

1. Predict first — Start best at first element 4; 2 does not replace it.
2. Model and explain — 9 is larger so best becomes 9; return 9.

Source model · final illustrated step

RETURN

best	**9**
result	**9**

Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict first
11 replaces initial best 8; neither 6 nor 10 exceeds 11.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Trace Largest([2,5,1]) and state initial and final best.

Question 2

Write pseudocode to return the minimum of a nonempty array values and explain its precondition.

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Evaluate what happens for Largest([-4,-2,-9]) and explain why initialising best=0 would be incorrect.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Initialising a maximum-tracking variable to 0 is correct for every integer array.

Use key vocabulary from memory

1. Explain algorithm in your own words.

2. Explain precondition in your own words.

3. Explain postcondition in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.